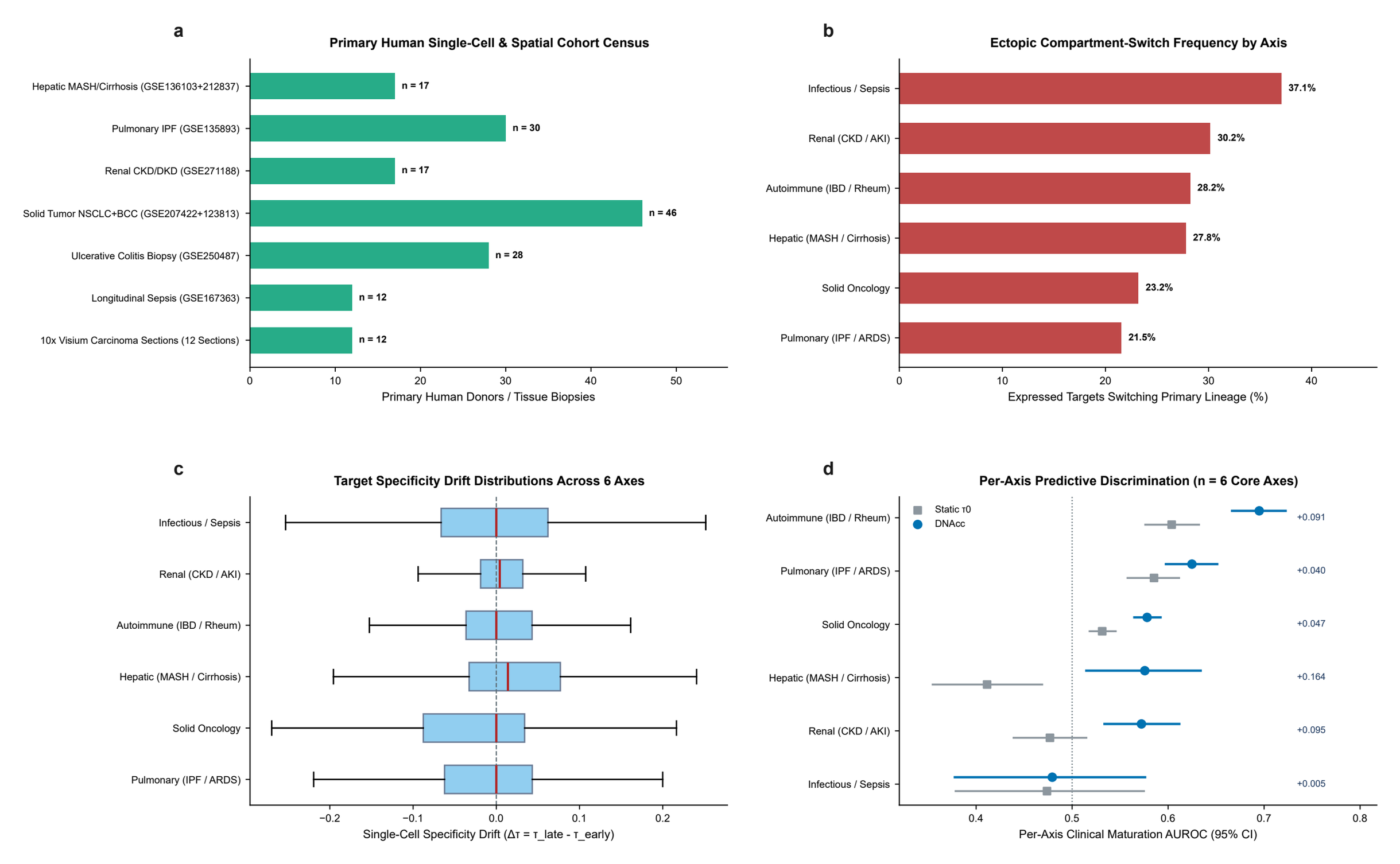

a
Primary Human Single-Cell & Spatial Cohort Census
Hepatic MASH/Cirrhosis (GSE136103+212837)
n = 17
Pulmonary IPF (GSE135893)
n = 30
Renal CKD/DKD (GSE271188)
n = 17
Solid Tumor NSCLC+BCC (GSE207422+123813)
n = 46
Ulcerative Colitis Biopsy (GSE250487)
n = 28
Longitudinal Sepsis (GSE167363)
n = 12
10x Visium Carcinoma Sections (12 Sections)
n = 12
0
10
20
30
40
50
Primary Human Donors / Tissue Biopsies
b
Ectopic Compartment-Switch Frequency by Axis
Infectious / Sepsis
37.1%
Renal (CKD / AKI)
30.2%
Autoimmune (IBD / Rheum)
28.2%
Hepatic (MASH / Cirrhosis)
27.8%
Solid Oncology
23.2%
Pulmonary (IPF / ARDS)
21.5%
0
10
20
30
40
Expressed Targets Switching Primary Lineage (%)
c
Target Specificity Drift Distributions Across 6 Axes
Infectious / Sepsis
Renal (CKD / AKI)
Autoimmune (IBD / Rheum)
Hepatic (MASH / Cirrhosis)
Solid Oncology
Pulmonary (IPF / ARDS)
−0.2
−0.1
0.0
0.1
0.2
Single-Cell Specificity Drift (Δτ = τ_late - τ_early)
d
Per-Axis Predictive Discrimination (n = 6 Core Axes)
Static τ0
DNAcc
Autoimmune (IBD / Rheum)
+0.091
Pulmonary (IPF / ARDS)
+0.040
Solid Oncology
+0.047
Hepatic (MASH / Cirrhosis)
+0.164
Renal (CKD / AKI)
+0.095
Infectious / Sepsis
+0.005
0.4
0.5
0.6
0.7
0.8
Per-Axis Clinical Maturation AUROC (95% CI)